

Liquid-Liquid Extraction

Final Lab Report
Unit Operations Lab 6
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1. Executive Summary (10 pts)

The objective of this experiment was to evaluate the performance of a three-stage countercurrent liquid liquid extraction system for the removal of acetic acid from a mineral oil phase using water as the solvent, and to assess the impact of agitator speed on mass transfer efficiency. The system was operated at flow rates of approximately 0.34 GPM for the oil phase and 0.67 GPM for the aqueous phase, corresponding to an operating ratio of $L/V=0.51$.

Acetic acid concentrations in both phases were determined via titration with 0.05 M NaOH. The measured data were used to construct equilibrium and operating lines and to perform a McCabe–Thiele analysis. The analysis, based on a feed composition of approximately 1.00 wt% acetic acid, predicted a requirement of three theoretical stages, which is consistent with the number of stages in the system.

Experimental results indicate that increased agitator speed enhances extraction performance. At 450 RPM, raffinate concentration decreased from 0.318 wt% to 0.177 wt% across three stages, while at 850 RPM it decreased from 0.212 wt% to 0.079 wt%, corresponding to a 62.7% reduction. However, calculated Murphree efficiencies ranged from 50.9% to 175%, with values exceeding theoretical limits indicating measurement uncertainty.

The system demonstrated effective acetic acid removal and behavior consistent with extraction theory. However, the reliability of efficiency calculations was limited by experimental error, particularly in concentration measurements. While the stage analysis was consistent with design expectations, improved analytical precision is required for accurate performance evaluation.

2. Introduction (10 pts)

Liquid-liquid or solvent extraction is a commonly used separation method in chemical engineering that takes advantage of the different in a solute's solubility between two immiscible liquid phases. Instead of solely relying on the vapor-liquid equilibrium like distillation, separation through extraction is done by touch a feed liquid with another immiscible solvent, causing the solute to be shared between the two phases [1]. Due to its effectiveness at operating at an ambient pressure and temperature, it is more energy-efficient alternative technique to distillation especially when the feed mixture forms an azeotrope, or the component that will be separated is at a low concentration [2].

Solvent extraction is used throughout different industries. In environmental engineering, it removes contaminants from wastewater. In the pharmaceutical industry, it's used to isolate then purify ingredients from complex reaction mixtures. In refining, it separates, hydrocarbons from aliphatic streams. In the food and biochemical industries, it recovers fermentation products like organic acids [3,4]. In industry a familiar operation is the

recovery of dilute acetic acid from aqueous streams. As the aqueous feed contains less than 30% acetic acid, liquid-liquid distillation is more economical than distillation due to the energy cost of evaporating water is bigger than the capital cost of the extraction equipment.

The main goal of the experiment is to operate a multi-stage mixer unit extracting acetic acid from white mineral oil with water as the solvent and analyzing how the agitator speed affects the extraction efficiency and the number of theoretical stages needed [1].

3. Theory (10 pts)

The main foundation of liquid-liquid extraction is phase equilibrium. As the acetic acid (solute) contacts two immiscible liquids where the white oil is the raffinate phase L and the water as the extract phase V. The acetic acid distributes itself between the water and acetic acid based on its relative solubility in each. If the concentrations are dilute enough where the two solvents stay immiscible, the equilibrium distribution is characterized through this relation [2].

Where w is the acetic acid weight fraction in the raffinate phase, v is the weight fraction in the extract phase and K is the equilibrium distribution coefficient. A value of $K > 1$ indicates the acetic acid stays in the oil. The goal of using the water as the solvent is to push it to the water side meaning that $K < 1$ or where enough solvent is used to overcome a bad K value [2][3].

For a cascade of n ideal stages with constant flow rates, solving the mass balances and equilibrium relations for each stage yields a closed expression. Instead, of working through each stage, the acetic acid concentration $n + 1$ could be written as a function of the concentration leaving stage 1 seen below [2]:

The E in this formula is the extraction factor which is defined as:

Here L is the oil (raffinate) flow rate and V is the water (extract) flow rate. E catches the competition between the flow rate (L/V) and equilibrium thermodynamics (K). A higher E means the system is positioned to move the solute to the extract phase. During operation, the solvent-to-feed ratio is then set so the extraction factor is greater than 1 guaranteeing that more extraction is possible with more stages [2][4].

Instead of using the above formula, the number of theoretical stages can be determined using the McCabe-Thiele method. For the dilute solutes in a countercurrent process, both phases flow rates are constant through stage to stage, so the operating line is straight-line on a y vs x diagram. The operating line is drawn between the feed end and the solvent end. The number of ideal stages is found through the stepping staircase method between the two lines (equilibrium curve and operating line) [1][2].

The actual mixer-settler stages don't reach perfect equilibrium as the mixing and residence times are limited and the phases separate before they fully reach equilibrium.

Liquid-liquid extraction stages stray from equilibrium because of insufficient mixing which makes efficiency modeling important for analyzing the performance of the overall process. The Murphree efficiency gives a stage-by-stage measure how close each stage is to the ideal. The extract phase at stage n is [1,2]:

Here y_n is the extract composition that would be in equilibrium with the raffinate leaving that stage. The numerator is the concentration change that happened whilst the denominator is the maximum change that almost happened if the stage were ideal. The expression for the raffinate phase below [1,2]:

Here x_n is the raffinate composition in equilibrium with the extract leaving the stage. The Murphree efficiency is stated for each stage as the change in concentration divided by the change predicted by the equilibrium. An efficiency of 1 means the stage has ideal performance and any value below 1 shows insufficient mixing or contact time. [2].

The overall column efficiency compares the theoretical stages needed against the actual available stages represented below [1]:

Here N is the number of required theoretical stages and n is the number of actual stages used. Throughout the experiment, the agitator speed is varied across three runs at constant flow rates to observe how mixing can affect the Murphree efficiency at each stage and the overall column efficiency.

4. Results (15 pts)

The purpose of this lab was to determine the mass transfer capabilities of a three-phase system involving mineral oil as the raffinate, water as the extract, and acetic acid as the solute. To allow for equilibrium mass transfer, the two phases are allowed to flow counter-currently past each other in three mixer-settler units that are connected in series. This flow regime allows for strong interaction between the phases, while agitators submerged within the mixture of each mixer-settler rotate and break apart the separate phases into smaller portions creating more surface area and a greater driving force for the diffusion of acetic acid to take place.

The counter-current flowrates of the two phases stayed constant throughout the duration of the lab. Although, the agitator speed was altered to determine its effects on mass transfer. At each rpm, the apparatus was allowed to reach a steady state, which was followed by the extraction of a sample from both phases, for each stage, that were then titrated with 0.05 NaOH to determine acetic acid concentration. This concentration data, in weight percent, along with the data in appendix I of the lab manual [1] was used to construct graphs, equilibrium diagrams, a McCabe-Thiele analysis, Murphree efficiencies, and an overall efficiency.

Acetic Acid Concentration vs. Stage Number & Agitator Speed:

The three speeds chosen for this lab were 450, 650, and 850 rpm. At every speed there was a notable trend where the final stage would see the lowest acetic acid concentration in both phases. This makes sense since a big portion of the solute has already been transferred to the extract phase in the first two stages. This leaves a small amount of solute available for mass transfer in the final stage.

It was also noticed that the solute concentrations for both phases were slightly lower at each subsequent agitator speed for all three stages. This is expected for the mineral oil phase since the increased agitator speed would cause for greater better mixing with an overall reduction in resistance to mass transfer, but not for the water phase. It seems that there could've have been consistent error when extracting samples of the water phases, allowing for samples having a considerable amount of mineral oil affecting titration and concentration results. Or it could be that the increase in agitator speed caused had adverse effects for this lab, creating an impediment to mass transfer rather than making it easier. Below are the graphs of acetic acid concentration vs mixer-settler stage and agitator rpm.

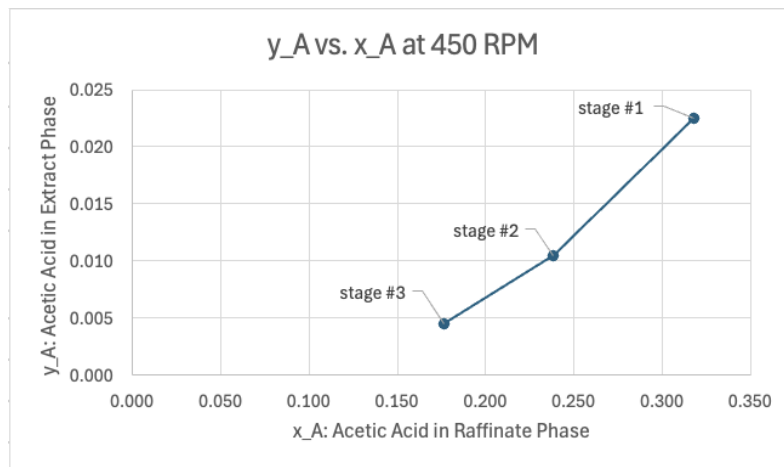


Figure 1. Acetic Acid Concentrations vs. 450 RPM

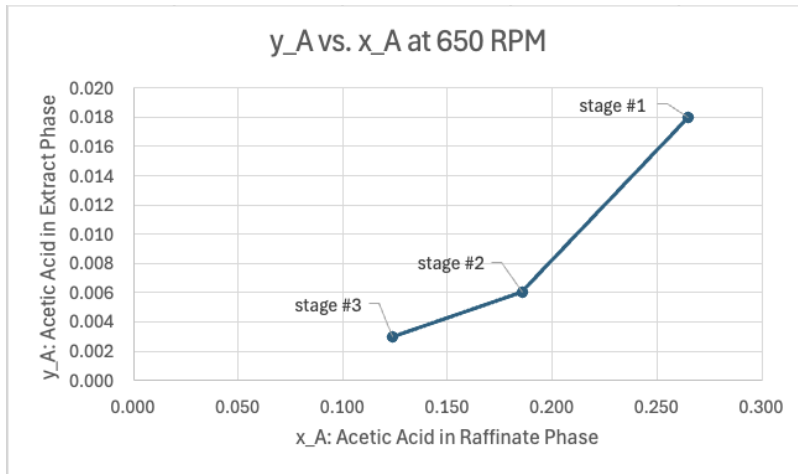


Figure 2. Acetic Acid Concentrations vs. 650 RPM

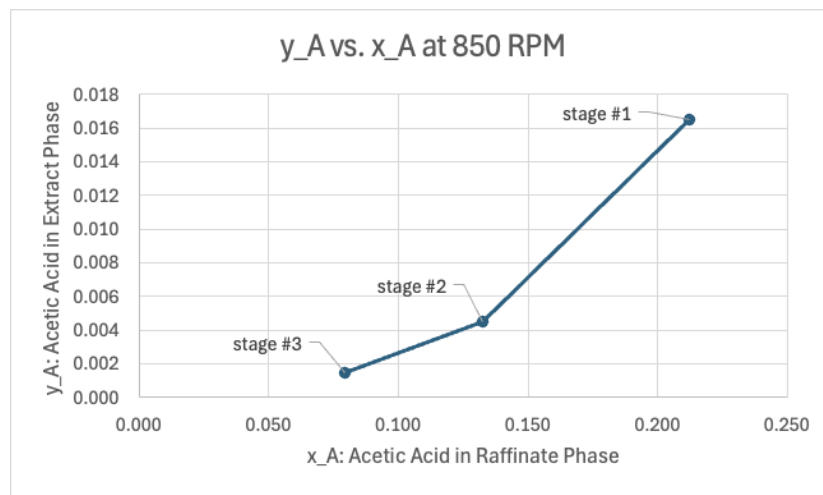


Figure 3. Acetic Acid Concentrations vs. 850 RPM

Murphree Efficiencies:

For the extract and the raffinate, the efficiencies were calculated by taking the average equilibrium constant, K , for each agitator RPM. Then, using the Murphree efficiency equations in [1] for both phases, the efficiencies were calculated for stages one and two at each speed. Stage three's efficiency couldn't be calculated because samples of the product tank were not taken and titrated to determine concentrations during the lab.

For the extract phase, the highest efficiency, 175%, was seen in stage 2 at a speed of 650 rpm; while the lowest efficiency, 92.9%, was seen in stage 1 at a speed of 450 rpm. Overall, there was no clear indicator of what rpm gave the best efficiency nor was there a certain stage. The apparatus as a whole showed very well efficiency across the board for the water phase. Below are tables of the data. Sample calculations for this data will be provided in the appendix section.

Table 1. Extract Phase Murphree Efficiencies

EXTRACT (450 RPM)		EXTRACT (650 RPM)		EXTRACT (850 RPM)	
stage #1		stage #1		stage #1	
K_avg =	23.62	K_avg =	28.96	K_avg =	26.12
y_n =	0.023	y_n =	0.018	y_n =	0.017
y*_n+1 =	0.010	y*_n+1 =	0.006	y*_n+1 =	0.005
y_n+1 =	0.011	y_n+1 =	0.006	y_n+1 =	0.005
E_ME,1 =	0.929	E_ME,1 =	1.03	E_ME,1 =	1.00
stage #2		stage #2		stage #2	
K_avg =	23.62	K_avg =	28.96	K_avg =	26.12
y_n =	0.011	y_n =	0.006	y_n =	0.005
y*_n+1 =	0.007	y*_n+1 =	0.004	y*_n+1 =	0.003
y_n+1 =	0.005	y_n+1 =	0.003	y_n+1 =	0.002
E_ME,2 =	1.71	E_ME,2 =	1.75	E_ME,2 =	1.52

For the raffinate phase, the highest efficiency calculated, 138%, was seen in stage 1 at a speed of 450 rpm. While the lowest, 50.9%, was found in stage 2 at 450 rpm. The range is much different for the mineral oil compared to the water. This indicates that the oil phase didn't reach an equilibrium mass transfer with the solute as easily as the water. This might indicate that the mass transfer coefficient is higher for the diffusion of acetic acid into water compared to mineral oil.

Table 2. Raffinate Phase Murphree Efficiencies

Raffinate (450 RPM)		Raffinate (650 RPM)		Raffinate (850 RPM)	
stage #1		stage #1		stage #1	
K_avg =	23.62	K_avg =	28.96	K_avg =	26.12
x_n =	0.318	x_n =	0.265	x_n =	0.212
x*_n+1 =	0.25982	x*_n+1 =	0.17376	x*_n+1 =	0.1306
x_n+1 =	0.238	x_n+1 =	0.185	x_n+1 =	0.132
E_MR,1 =	1.38	E_MR,1 =	0.877	E_MR,1 =	0.983
stage #2		stage #2		stage #2	
K_avg =	23.62	K_avg =	28.96	K_avg =	26.12
x_n =	0.238	x_n =	0.185	x_n =	0.132
x*_n+1 =	0.1181	x*_n+1 =	0.08688	x*_n+1 =	0.05224
x_n+1 =	0.177	x_n+1 =	0.124	x_n+1 =	0.079
E_MR,2 =	0.509	E_MR,2 =	0.622	E_MR,2 =	0.664

Overall, there was no clear trend on the effect of agitator speed on the Murphree efficiencies. Within each phase, the numbers were all over the place no matter what the speed or the stage was. Although, from the numbers it can be noted that for the water phase, efficiency is consistently higher for stage two compared to stage one regardless of the speed. For the mineral oil phase it's the opposite, the efficiency is consistently higher for stage one compared to stage two. It seems that these trends might arise from the overall characteristics of the apparatus and experimental set-up rather than the inherent mass transfer behavior of the phases involved.

McCabe-Thiele Analysis:

The McCabe–Thiele method is used to analyze staged separation processes for liquid–liquid extraction systems. In this experiment, the method was applied to evaluate the extraction of acetic acid from a mineral oil phase using water as the solvent. The analysis is based on the distribution of the solute between two immiscible phases and the relationship between equilibrium and operating conditions. By combining equilibrium data with mass balance considerations, the McCabe–Thiele approach allows for the determination of the number of theoretical stages required to achieve a desired separation.

To perform this analysis, the feed composition was first determined experimentally through titration using a standardized sodium hydroxide solution. From this, the concentration of acetic acid in the oil phase was calculated and used as the starting point for graphical construction. The equilibrium relationship between the raffinate and extract phases was obtained from tabulated data, while the operating line was defined based on the ratio of the flow rates of the two phases. A McCabe Thiele diagram was constructed, and stage stepping was performed to estimate the number of theoretical stages required for the extraction process.

This analysis provides insight into the efficiency of the extraction system and allows for comparison between the theoretical performance and the actual number of stages available in the experimental setup.

To begin the McCabe Thiele analysis, the first step necessary to determine the composition of the feed stream, to serve as the starting point for graphical construction. The feed composition used in the McCabe–Thiele analysis comes from titration data using 0.05 M NaOH, with a measured volume of 56.6 mL. From this, the number of moles of acetic acid was calculated to be 0.00283 mol, corresponding to a mass of 0.169942 g of acetic acid in the sample. Using the sample volume of 0.02 L and a density of 850 g/L, the total sample mass was determined to be 17 g. This resulted in a feed composition of $X_F = 0.999656\%$, which is simplified to approximately 1.00 wt% acetic acid. This value is consistent with the expected feed concentration for the experiment and provides a realistic basis for the McCabe–Thiele analysis. The operating condition, defined by the ratio of raffinate to extract flow rates, was calculated as $L/V = 0.51$, which determines the slope of the operating line. (Refer to Table 3 and 4 and sample calculations for value)

Table 3: Given Values needed for McCabe Thiele plot

Given Data	
Concentration	Description
0.05	NaOH M
56.6	NaOH Vol (mL)
0.02	Sample Vol (L)
850	Density (g/L)
60.05	MW of Acid

Table 4: Calculated values for McCabe Thiele plot

Values needed to Calculate for McCabe Thiele	
0.00283	NaOH mol
0.16994	Mass Of Acid
2	
17	Mass Sample
0.99965	Xf
6	
0.51	L/V ratio

With parameters and values prepared the McCabe Thiele plot can be constructed from equilibrium data, and operating line is shown in Figure 4. The equilibrium curve from data plots in Table 5 [1] exhibit a nonlinear increase in acetic acid concentration in the extract phase as the raffinate concentration increases, indicating a strong preference for the solute to partition into the water phase. In contrast, the operating line [3] from data in Table 6 is nearly horizontal due to the relatively low L/V ratio of 0.51, reflecting the operating conditions of the extraction process. The significant vertical separation between the equilibrium curve and the operating line demonstrates a strong driving force for mass transfer throughout the system.

Stagewise stepping was performed on the plot by alternating vertical movements to the equilibrium curve and horizontal movements to the operating line. As shown in the staircase construction, the first stage results in a substantial decrease in raffinate concentration, indicating that a large fraction of acetic acid is removed in

the initial contact. Subsequent stages produce smaller changes as the system approaches equilibrium. After approximately three steps, the staircase reaches the lower concentration region, indicating that three theoretical stages are required for separation. This result is consistent with the expected behavior of the system and confirms that the extraction process is highly efficient, with most of the solute removal occurring in the first few stages. Thus, conducting that the theoretical number of stages needed for separation should be three which coincide with the apparatus having three separation units.

Table 5: Equilibrium data for McCabe Thiele plot

Equilibrium Data	
0	0.26
0.02	0.9
0.03	1.19
0.05	1.79
0.06	2.39
0.07	3.27

Table 6: Operating line values for McCabe Thiele plot

Operating Line	
0	0
0.02	0.0102
0.03	0.0153
0.05	0.0255
0.06	0.0306
0.07	0.0357

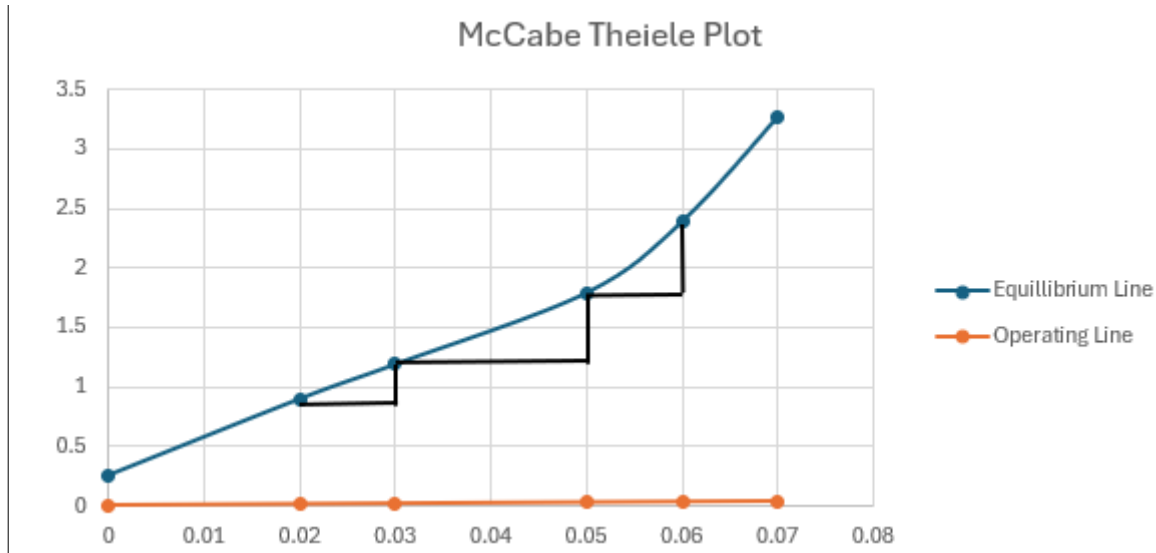


Figure 4. Operating Line Values for McCabe-Thiele Plot

Overall Efficiency & Efficiency:

With the theoretical number of stages needed equal to three, the efficiency calculation found in [1] is simple and straight-forward give a value of $\eta = 1.0$. This is the same result for the overall efficiency, , found within the same source.

5. Discussion (15 pts)

The experimental results show partial agreement with liquid-liquid extraction theory, particularly in terms of solute distribution and stage performance. The equilibrium data demonstrates a strong preference for acetic acid to partition into the aqueous phase, as the extract concentration increases from $Y=0.26$ at $X=0$ to $Y=3.27$ at $X=0.07$. This nonlinear relationship is consistent with equilibrium-stage theory and confirms a favorable thermodynamic driving force for extraction [2]. The operating line, defined by $L/V=0.51$, is relatively shallow, resulting in a large separation from the equilibrium curve and further supporting efficient mass transfer.

The McCabe Thiele analysis, based on a feed composition of XF being approximately $\approx 1.00\%$, predicts approximately three theoretical stages. This matches the three-stage configuration of the mixer-settler system and suggests that the process is operating near ideal stage requirements. However, this agreement should be interpreted cautiously, as the graphical method is sensitive to the limited equilibrium data range ($X = 0$ to 0.07) and may reinforce this result according to the sensitivity of McCabe Thiele graphs [4].

The raffinate-phase concentration data shows trends that align with theoretical expectations. At 450 RPM, acetic acid concentration decreases from 0.318 wt% in stage 1 to 0.177 wt% in stage 3 (44.3% reduction), while at 850 RPM it decreases from 0.212 wt

% to 0.079 wt% (62.7% reduction). This indicates that increased agitator speed improves extraction performance, consistent with enhanced interfacial area and reduced mass transfer resistance [3]. Additionally, the final-stage concentration decreases by approximately 55% when increasing speed from 450 RPM to 850 RPM, further supporting improved mass transfer at higher agitation.

In contrast, the Murphree efficiency results show significant deviation from theoretical expectations. Extract phase efficiencies range from 92.9% to 175%, while raffinate-phase efficiencies range from 50.9% to 138%. Values exceeding 100% are not physically possible and indicate systematic experimental error rather than true process performance [1][2]. The most significant source of error is likely titration inaccuracy, particularly endpoint overshoot when using phenolphthalein. This effect is amplified at lower concentrations, where smaller titration volumes lead to higher relative uncertainty. For example, a ± 0.05 mL uncertainty corresponds to approximately 0.28% error at 18 mL but increases to approximately 1.1% at 4.5 mL, contributing to error propagation in efficiency calculations [4].

Overall, while the system exhibits expected qualitative behavior specifically decreasing solute concentration with increasing stage number and agitation, the quantitative accuracy of the results is limited by measurement uncertainty. The agreement between theoretical and actual stages is therefore not sufficient alone to validate the model, as it is influenced by data constraints and experimental error.

In conclusion, the experiment supports the fundamental principles of liquid-liquid extraction and demonstrates effective removal of acetic acid; however, significant deviations in efficiency calculations highlight limitations in experimental precision. These results emphasize the importance of accurate analytical methods and controlled operating conditions when applying theoretical models such as the McCabe Thiele method.

6. References (10 pts)

- [1] *Liquid-Liquid Extraction*. ChE 382 Unit Operations Laboratory Manual, Spring 2025v1.
- [2] McCabe, W. L.; Smith, J. C.; Harriott, P. *Unit Operations of Chemical Engineering*, 7th ed.; McGraw-Hill: New York, 2005.
- [3] King, C. J. *Separation Processes*, 2nd ed.; McGraw-Hill: New York, 1980.
- [4] Wankat, P. C. *Equilibrium-Staged Separations*; Prentice Hall: Englewood Cliffs, NJ, 1988.

[5] Koch Modular Process Systems. Typical Industrial Applications for Separation by Liquid-Liquid Extraction. <https://kochmodular.com/liquid-liquid-extraction/industrial-applications/> (accessed April 2026).

[6] Mousavi, S. et al. Redefining Stage Efficiency in Liquid-Liquid Extraction: Development and Application of a Modified Murphree Efficiency. *PSE Community*, 2025. <https://psecommunity.org/LAPSE:2025.0373>

7. Appendix I: Data Tabulation/Graphs (10 pts)

Table 7: Mineral Oil Sample Parameters at 450 RPM

Mixer-Settler Stage	NaOH Needed (mL)	[Acetic Acid] (mol/L)	Acetic Acid wt%
1	18.0	0.045	0.318
2	13.5	0.034	0.238
3	10.0	0.025	0.177

Table 8: Mineral Oil Sample Parameters at 650 RPM

Mixer-Settler Stage	NaOH Needed (mL)	[Acetic Acid] (mol/L)	Acetic Acid wt%
1	15.0	0.038	0.265
2	10.5	0.026	0.185
3	7.0	0.018	0.124

Table 9: Mineral Oil Sample Parameters at 850 RPM

Mixer-Settler Stage	NaOH Needed (mL)	[Acetic Acid] (mol/L)	Acetic Acid wt%
1	12.0	0.030	0.212
2	7.5	0.019	0.132
3	4.5	0.011	0.079

Table 10: Water Sample Parameters at 450 RPM

Mixer-Settler Stage	NaOH Needed (mL)	[Acetic Acid] (mol/L)	Acetic Acid wt%
1	1.5	0.00375	0.023

2	0.7	0.00175	0.011
3	0.3	0.00075	0.005

Table 11: Water Sample Parameters at 650 RPM

Mixer-Settler Stage	NaOH Needed (mL)	[Acetic Acid] (mol/L)	Acetic Acid wt%
1	1.2	0.0030	0.018
2	0.4	0.0010	0.006
3	0.2	0.0005	0.003

Table 12: Water Sample Parameters at 850 RPM

Mixer-Settler Stage	NaOH Needed (mL)	[Acetic Acid] (mol/L)	Acetic Acid wt%
1	1.1	0.00275	0.017
2	0.3	0.00075	0.005
3	0.1	0.00025	0.002

8. Appendix II: Error Analysis (10 pts)

For the quantitative error analysis, the main errors come from titration readings, flowrate changes, and measurements of the agitator speed during operation. As the acetic acid concentration was derived from the titration volumes, small reading errors affect the concentration values. For the titration volumes, a burette uncertainty of about + 0.05 mL is assumed, and this has a bigger effect on smaller titration volumes. For example, at 450 RPM Stage 1, 18 mL of NaOH was used which increased the relative error to 0.28%. However, at 850 RPM Stage 3, 4.5 mL was used bring the relative error to 1.1%. This indicates that lower concentration samples were more sensitive to measurement errors. Despite the uncertainty, the results showed some trends. At 450 RPM, the acetic acid in the oil phase dropped from 0.318 wt% in Stage 1 to 0.177 wt% in Stage 3 which is a 44.3% decrease. AT 850 RPM, it dropped from 0.212 wt% to 0.079 wt% showing a 62.7% decrease. When comparing the final stage 3 values, increasing the speed from 450 RPM to 850 RPM, reducing the remaining acetic acid by 55%. If the tachometer uncertainty was + 10 RPM which is the same 2.22% error at 450 RPM and 1.18% at 850 RPM. Any changes, even small in the water or feed flowrate can affect the solvent-to-feed ratio which changes the extraction efficiency between trial runs.

The main qualitative errors in this lab were due to mixing, steady-state conditions, phase separation and general operator technique. An issue is the incomplete separation of both liquid phases in the settlers. If small water droplets were present in the oil sample or vice versa, the compositions do not represent fully the equilibrium conditions. At higher

speeds for example 850 RPM, stronger mixing probably improved mass transfer but could have caused emulsions to take longer to separate. At lower speeds like 450 RPM, the mixing is too weak to fully separate the liquids, and this reduces interfacial area thereby reducing the extraction performance. Another source of error was the sampling before the system stabilized. The lab manual recommended allowing the unit to run until steady liquid levels are reached and the oil leaves to the product tank before taking any samples. Titration endpoint spotting also introduced error as the phenolphthalein color change depends on visual judgement. This was very important as the NaOH is more difficult to titrate with a non-aqueous phase and needs a lot of mixing for good and accurate data.

9. Appendix III: Sample Calculations (10 pts)

Initial Acetic Acid wt% in the Feed:

The density of the solution is taken to be the density of the white mineral oil

Calculating the Amount of Acetic Acid to Add to Solution:

Murphree Efficiencies:

**For our calculations we calculated the average K value for each agitator speed and used that value; for agitator speed 650 RPM, $K = 28.96$ **

Data from an agitator speed of 650 rpm in stage one was used for this sample calculation

Overall Efficiency and Efficiency:

Since our McCabe-Thiele analysis gave a theoretical number of stages equivalent to the actual number, the value of both efficiencies is equal to one; the calculation is trivial

McCabe Thiele Theoretical Stage Calculations

The following calculations are used for finding the moles of NaOH to find the mass of Acetic Acid in the feedstock of the apparatus

Mass Calculations for Acetic Acid Mass:

Calculations for the sample mineral oil (MO) that will be used for the McCabe Thiele plot for theoretical stages:

Calculating the operating line requires using equilibrium data and previously converted values to make an equation for the operating line in excel

General Operating line equation:

Summary of units in the equation

Unit	Application
	Same as the equilibrium data value for x
	Operating line equation
L	Oil Flow rate (GPM)
V	Water Flow rate (GPM)

Operating line equation can be simplified by turning the flow rates of oil and water into a fixed value

Flow Rate Ratio:

Operating line equation can be simplified into:

The following is a sample calculation of the first iteration of the operating line equation for the McCabe Thiele plot:

10. Appendix IV: Individual Team Contributions

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	6 hours	Joshua Laforest, Cesearo Dilosa, Javier Silva, Ify Bidokwu
Pre-Lab Editing	40 minutes	Joshua Laforest, Cesearo Dilosa, Javier Silva, Ify Bidokwu
Experimental Objective (Prelab)	20 minutes	Joshua Laforest
Apparatus (prelab)	45 minutes	Javier Silva

Materials & Supplies (prelab)	35 minutes	Ify Bidokwu, Cesearo Dilosa
Procedure (prelab)	1 hour	Cesearo Dilosa
Project Safety Evaluation (prelab)	45 minutes	Ify Bidokwu, Cesearo Dilosa
Final Lab Editing	1 hour	Joshua Laforest, Cesearo Dilosa, Javier Silva, Ify Bidokwu
Executive Summary (final lab)	50 minutes	Joshua Laforest
Introduction (Final lab)	40 minutes	Ify Bidokwu
Theory (Final Lab)	40 minutes	Ify Bidokwu
Results (final lab)	1.5 hours	Cesearo Dilosa
Discussion (final lab)	1.5 hours	Javier Silva
References (final and/or prelab)	20 minutes	Javier Silva, Cesearo Dilosa, Ify Bidowu, Joshua laforest
Data Tabulation/Graphs (final)	1.5 hours	Cesearo Dilosa, Joshua Laforest
Error Analysis (final lab)	50 minutes	Ify Bidokwu
Sample Calculations (final Lab)	1.5 hours	Cesearo Dilosa, Javier Silva
Power Point Presentation	45 minutes	Joshua Laforest, Cesearo Dilosa, Javier Silva, Ify Bidokwu
Total	21 hours	